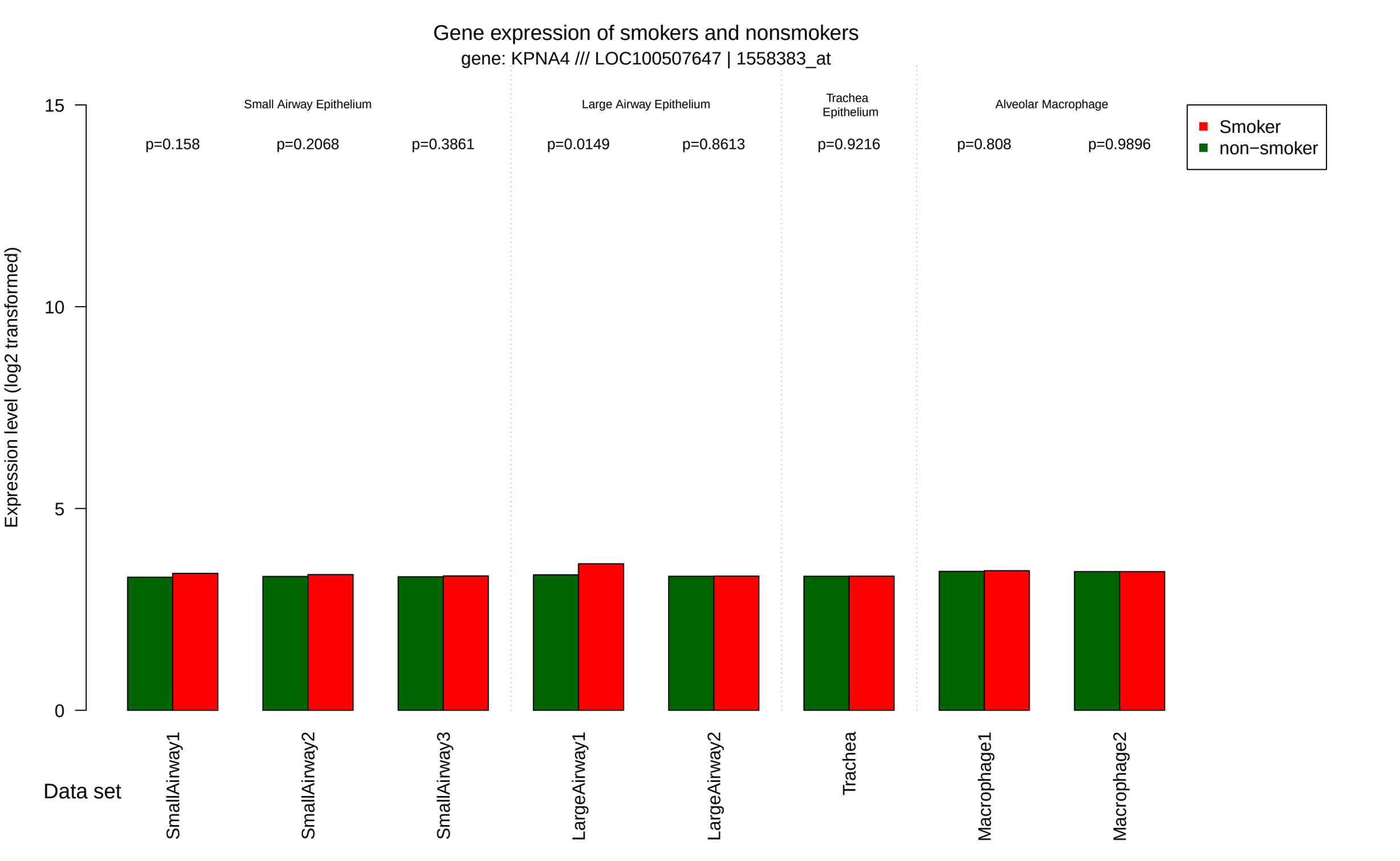
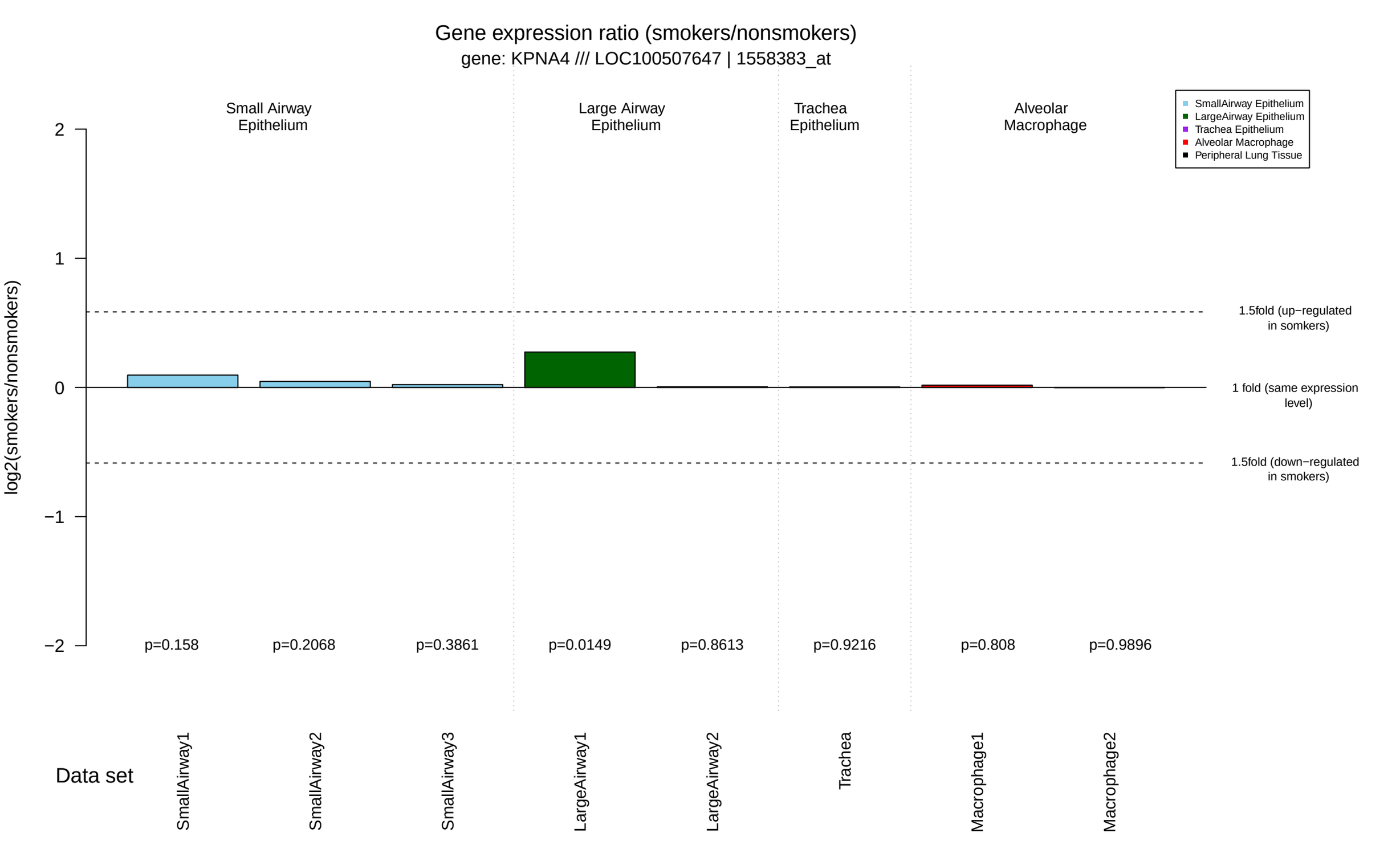
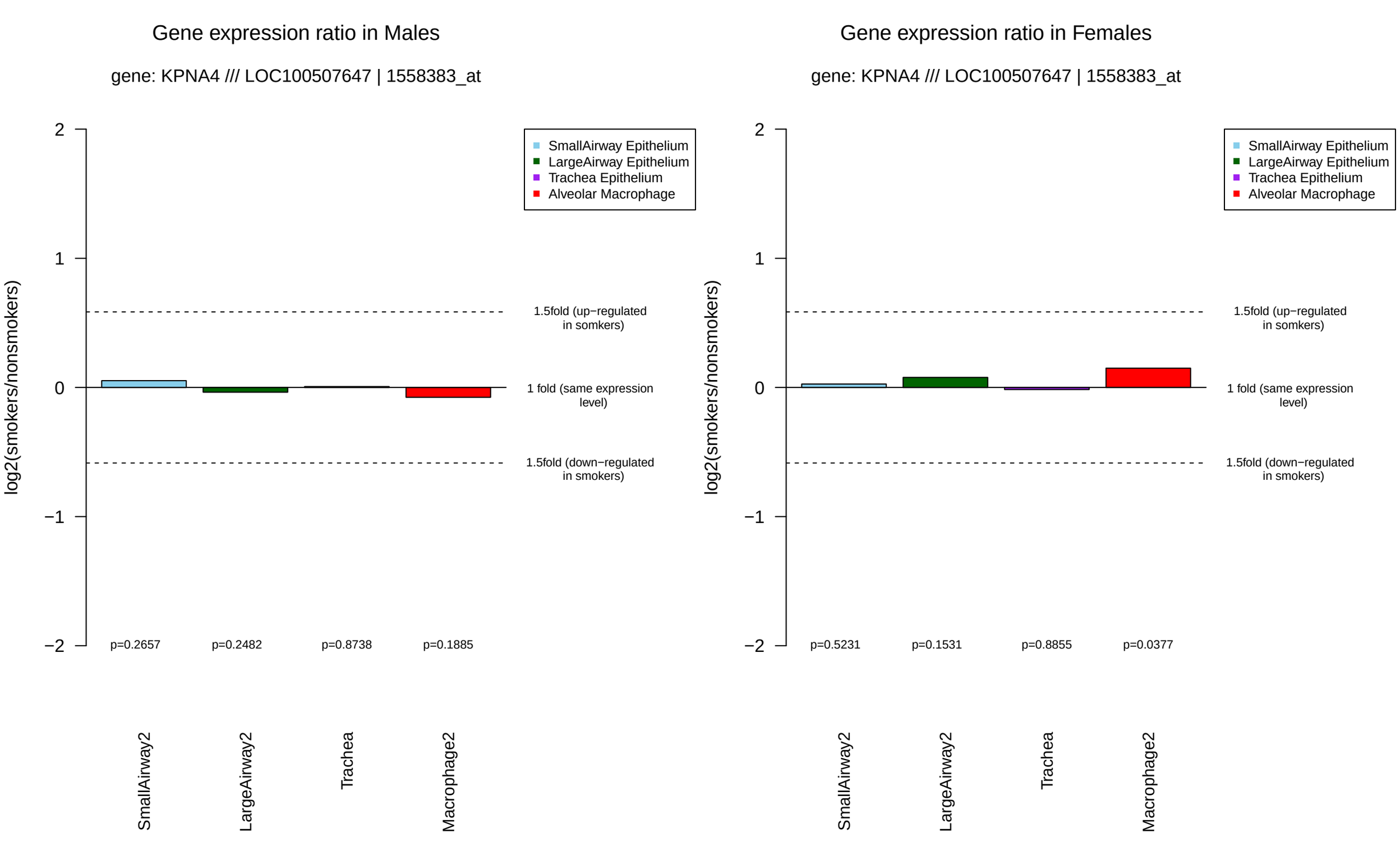
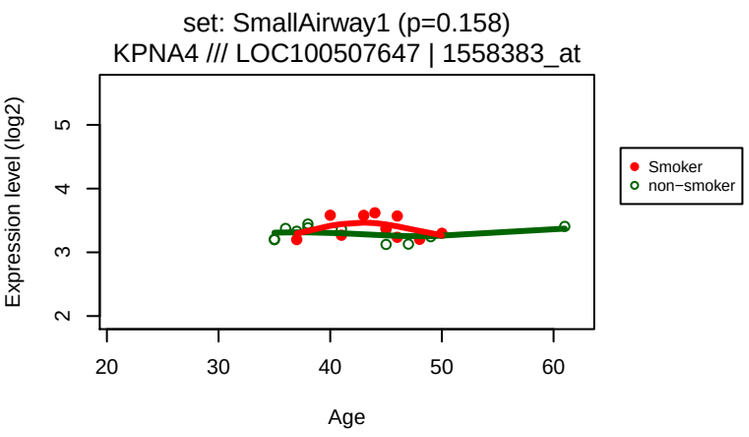
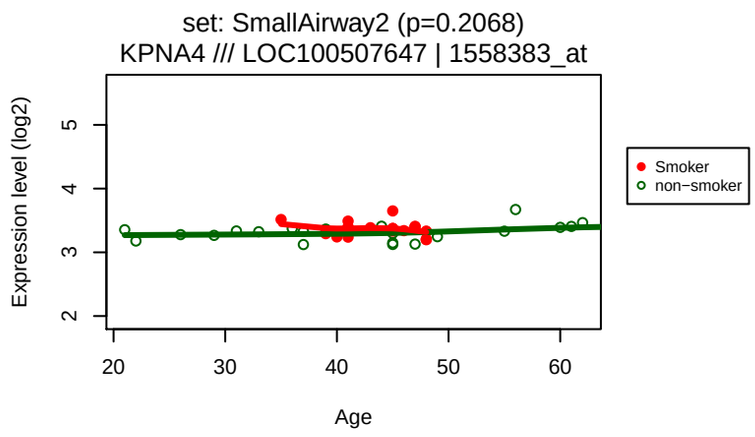
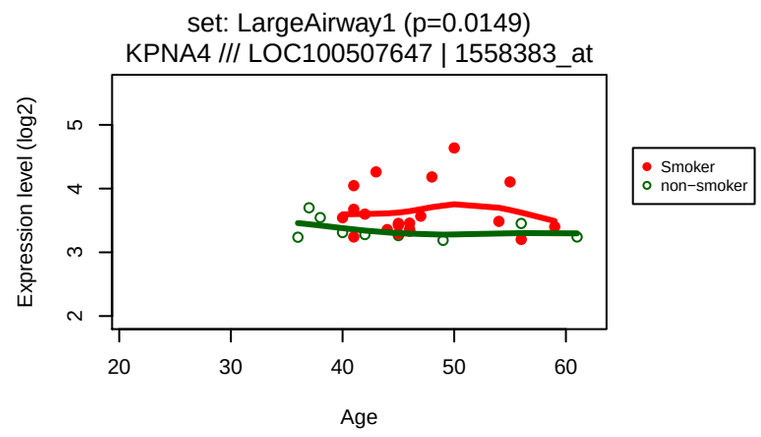

Gene expression of smokers and nonsmokers
gene: KPNA4 /// LOC100507647 | 1558383_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.158
p=0.2068
p=0.3861
p=0.0149
p=0.8613
p=0.9216
p=0.808
p=0.9896
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

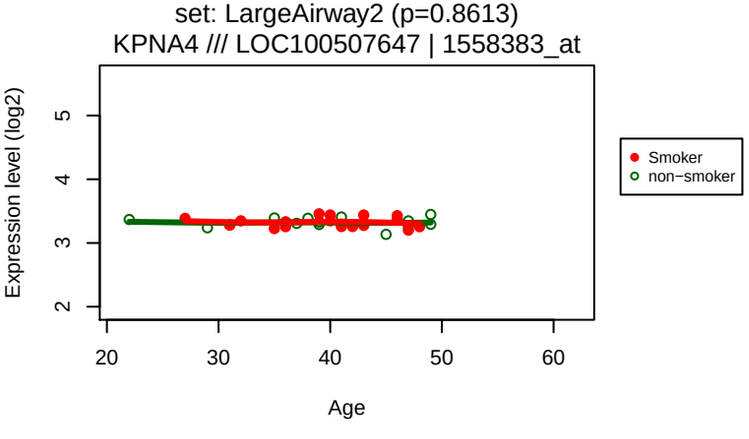

Gene expression ratio (smokers/nonsmokers)
gene: KPNA4 /// LOC100507647 | 1558383_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.158
p=0.2068
p=0.3861
p=0.0149
p=0.8613
p=0.9216
p=0.808
p=0.9896
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

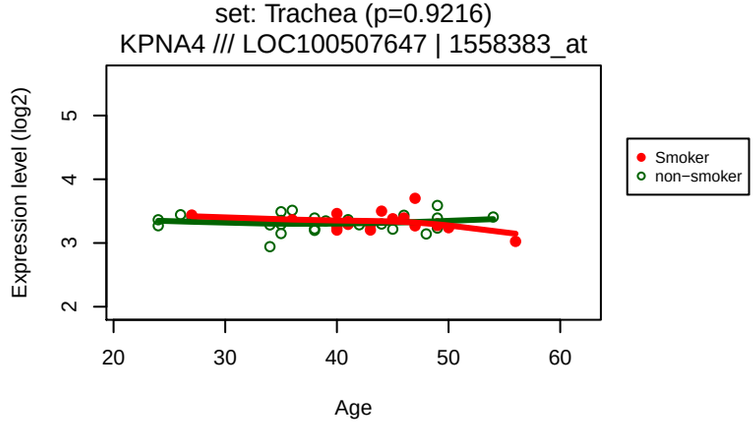

Gene expression ratio in Males
gene: KPNA4 /// LOC100507647 | 1558383_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2657
p=0.2482
p=0.8738
p=0.1885
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KPNA4 /// LOC100507647 | 1558383_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5231
p=0.1531
p=0.8855
p=0.0377
SmallAirway2
LargeAirway2
Trachea
Macrophage2

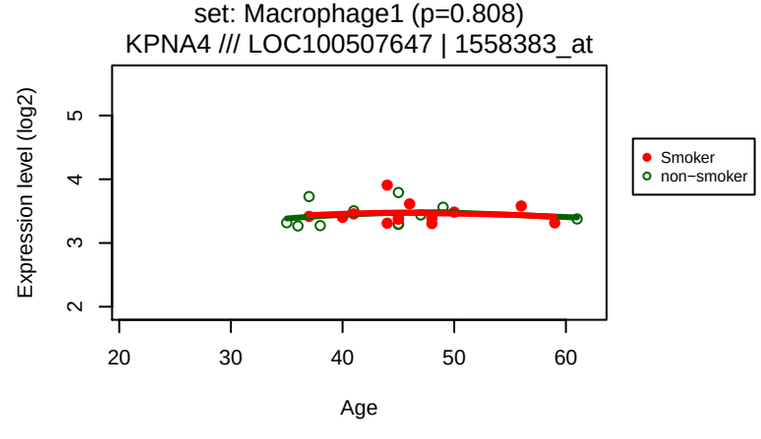

set: SmallAirway1 (p=0.158)
KPNA4 /// LOC100507647 | 1558383_at
Expression level (log2)
Age
Smoker
non−smoker

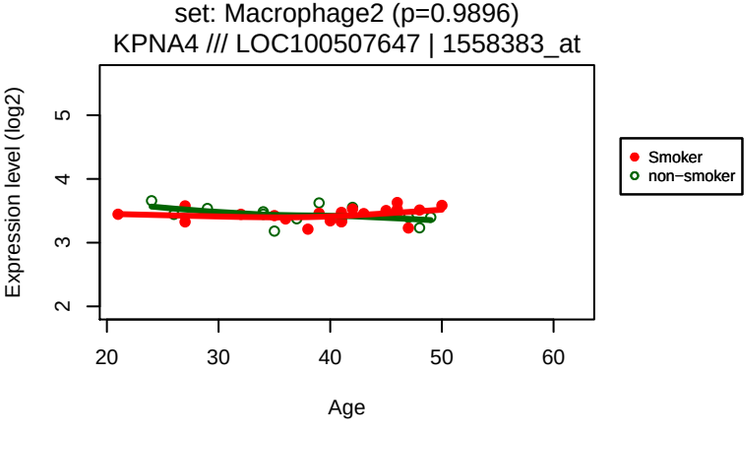

set: SmallAirway2 (p=0.2068)
KPNA4 /// LOC100507647 | 1558383_at
Expression level (log2)
Age
Smoker
non−smoker

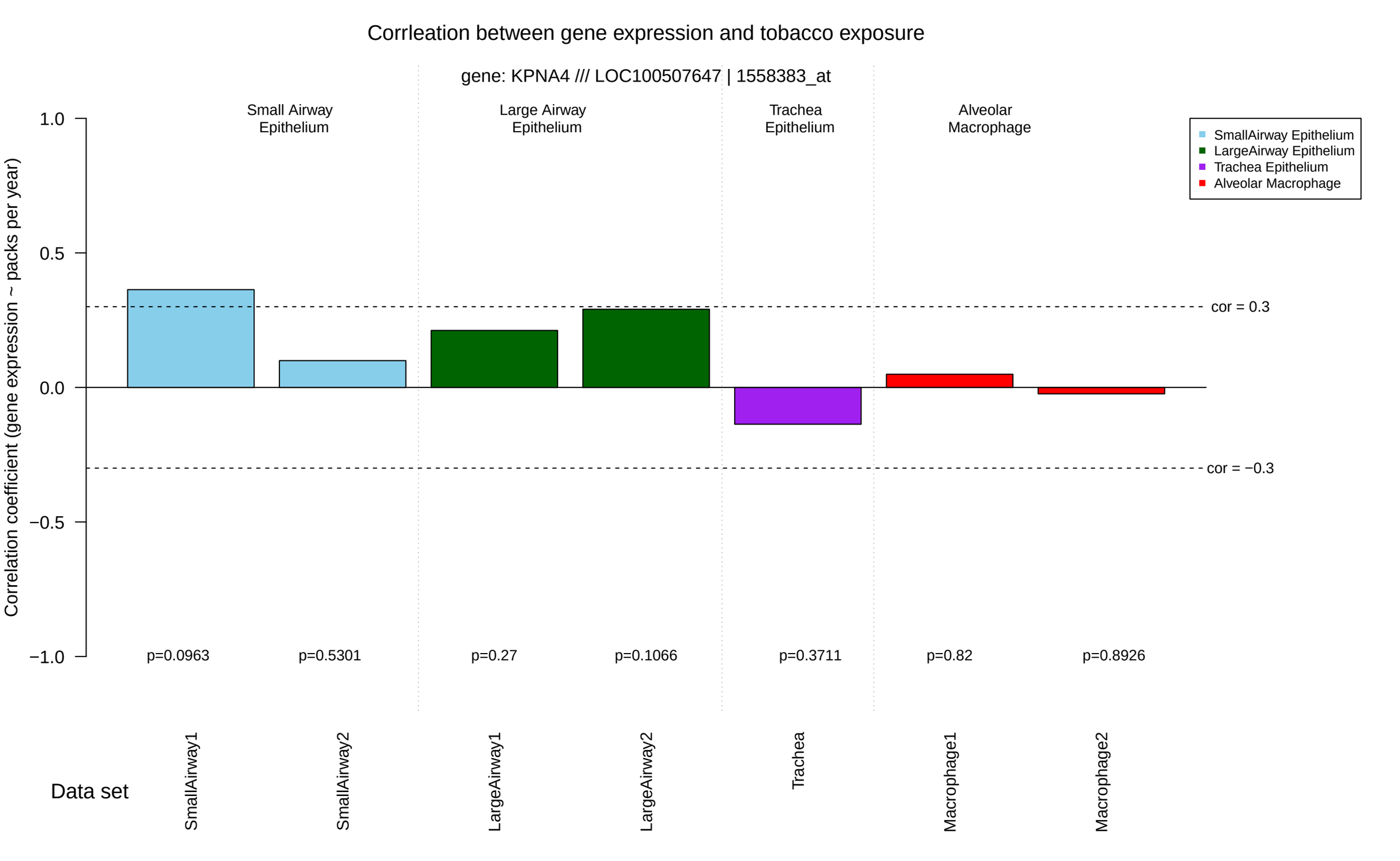

set: LargeAirway1 (p=0.0149)
KPNA4 /// LOC100507647 | 1558383_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8613)
KPNA4 /// LOC100507647 | 1558383_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9216)
KPNA4 /// LOC100507647 | 1558383_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.808)
KPNA4 /// LOC100507647 | 1558383_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9896)
KPNA4 /// LOC100507647 | 1558383_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KPNA4 /// LOC100507647 | 1558383_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0963
p=0.5301
p=0.27
p=0.1066
p=0.3711
p=0.82
p=0.8926
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2